European Developer Conference



Presented By:
Nathanial Freitas,
Senior Manager,
Java



Java on Tungsten through WebSphere

Palm Solutions Group
September 2003

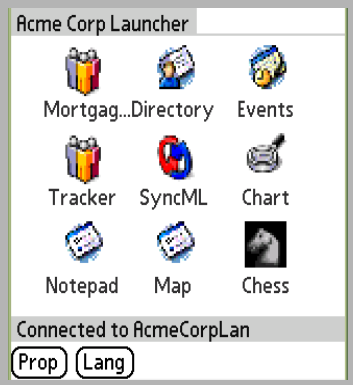
Java Now on Tungsten Handhelds

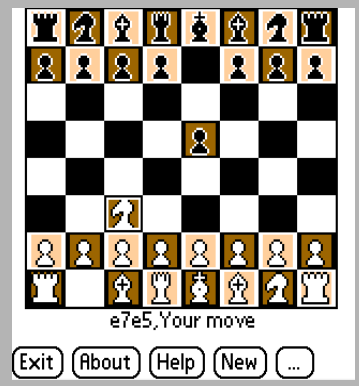
Tungsten handhelds now include IBM's WebSphere® Micro Environment, a Java (J2ME) software runtime

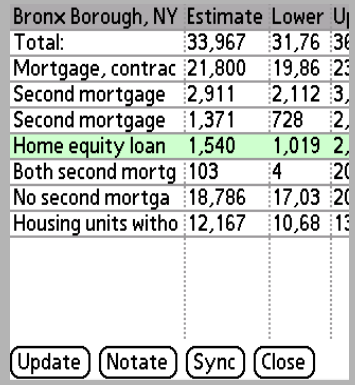
Java complements and enhances Palm OS platform- users can run J2ME applications, side by side with Palm OS applications!

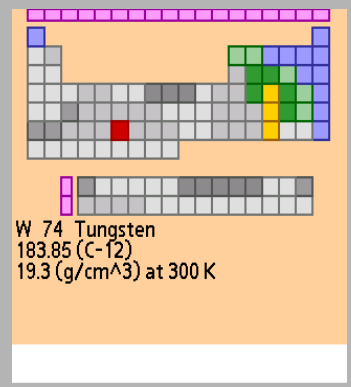
Tungsten developer community expands with more than 3 million Java developers, who can now leverage their tools and training

*thanks to Spruce Technologies and Markus Goldkamp











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Why Java?



STANDARDS: Java technology is on 100 percent of the standards lists of Fortune 500 companies

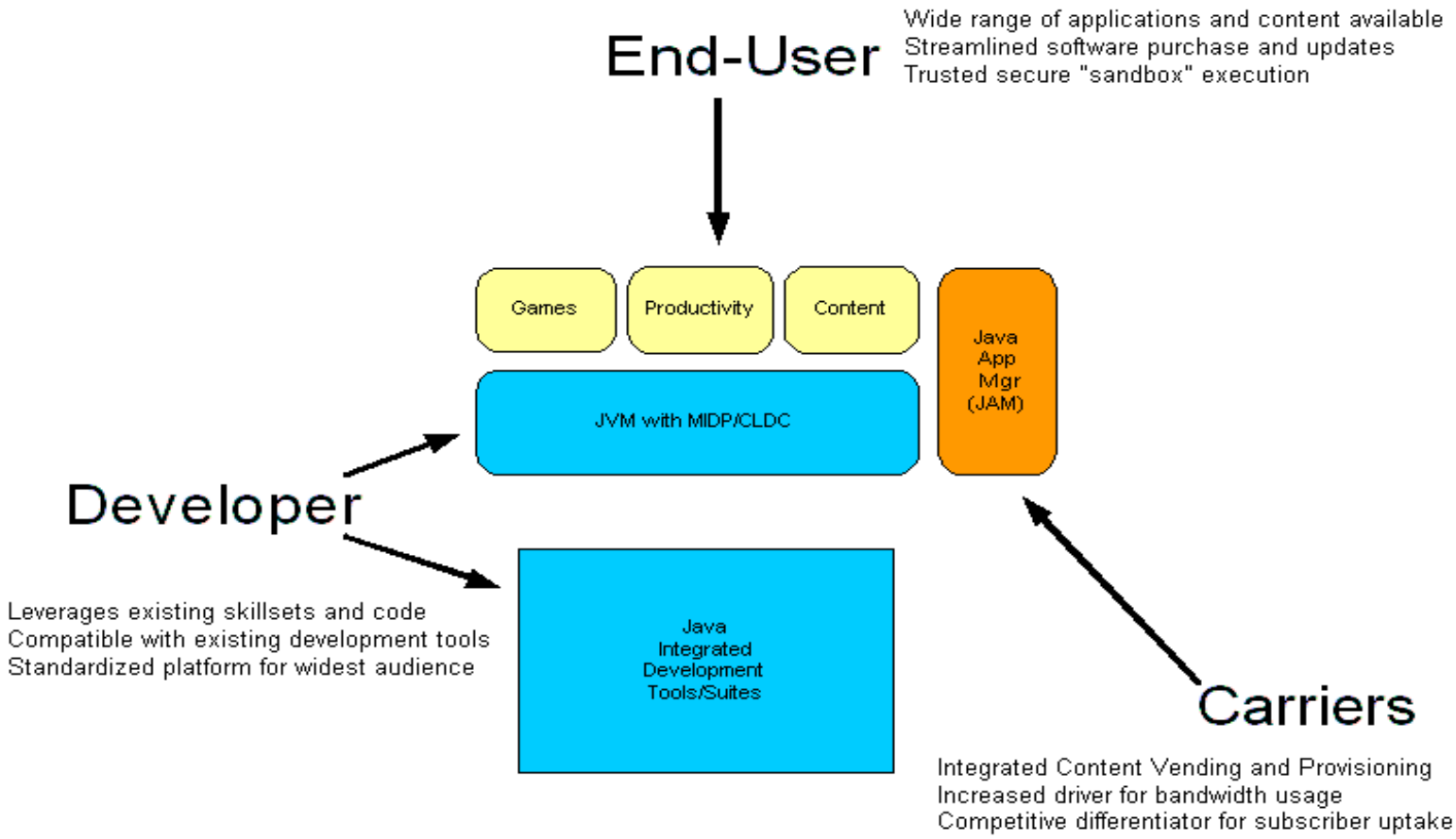
PLATFORM: 78 percent of executives view Java technology as the best platform for Web services (source: Giga, Computerworld).

LEVERAGED: More than 150 Java technology-enabled models are available from 22 manufacturers

CONNECTED: The Java platform is a leading platform for mobile application provisioning and services, with more than 53 carrier deployments and 11 trials around the world

*data from JavaOne 2003 Keynotes

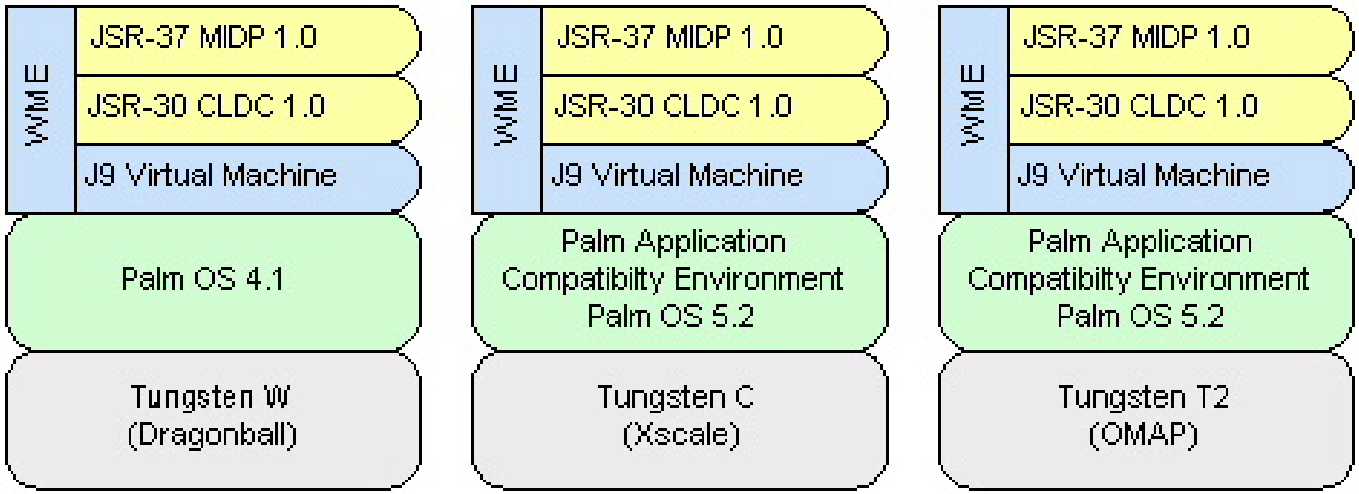
Benefits to Many Segments



Phase I Release – Now!

Initial release, CLDC 1.0 and MIDP 1.0 certified

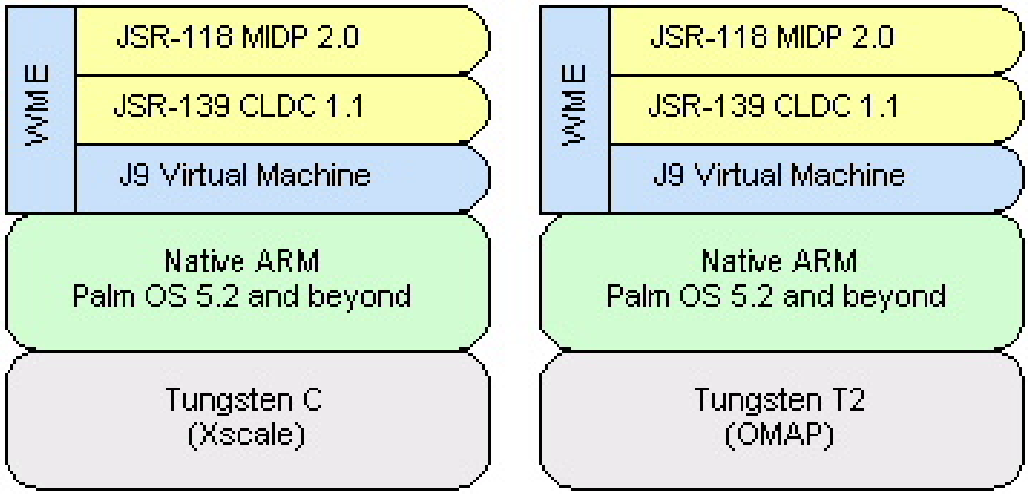
- Tungsten-enabled version of existing WebSphere Micro Environment; focus is on providing support for base Java Specifications (see jcp.org) and developer tools
- Transparent support for Palm OS 4.1 and Palm OS 5.2 (PACE), Tungsten W, C and T2 handhelds
- Available to developers from www.palm.com/java on to begin developing and testing Java apps on Tungsten handhelds



Phase II Release

Subsequent release will be CLDC 1.1 and MIDP 2.0 certified

- Support for Palm OS 5.2 (native ARM): Tungsten C, T2 and future handhelds
- Focus is on performance optimizations
- Will be broadly available for end users to deploy J2ME-based applications and solutions



Breaking Perceived Limitations of MIDP

Enhanced user experience with rich multimedia capabilities and high resolution screen, able to render complex tables, charts, and PNG images

Superior usability and options for data entry

- Five-way navigation, integrated keyboard, Graffiti® 2, and optional portable keyboards

Multiple wireless network support (Bluetooth, GPRS, 802.11b)

Greater memory and storage than most handsets. For example:

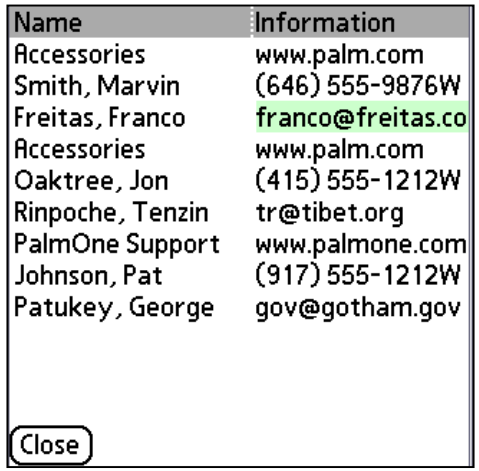
	Tungsten W	Tungsten C	Tungsten T2
Dynamic Heap	1 MB	4 MB	1 MB
Storage Heap	14.8 MB	51MB	29.5 MB

MIDlet application can launch from and utilize an external Secure Digital (SD) storage card (with sizes up to 1GB)

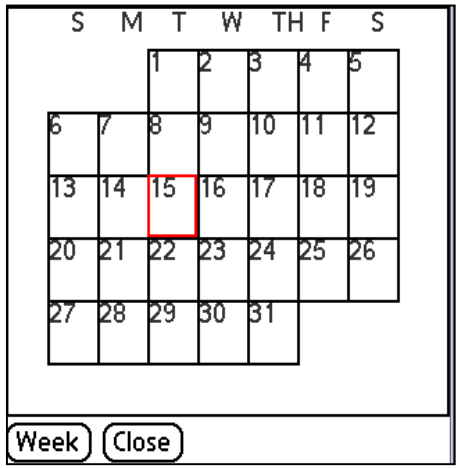
User Interface Parity Examples



Palm OS Launcher – in Java!



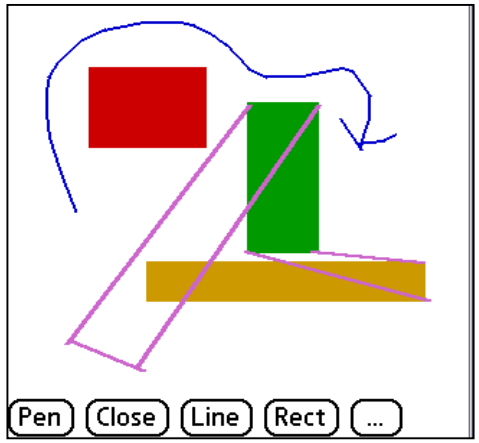
AddressBook Look-a-Like



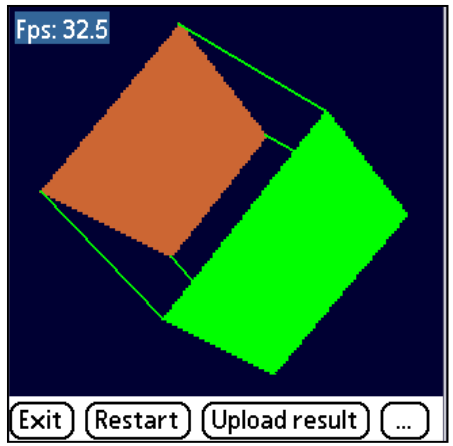
DateBook Look-a-Like



Survey Form built with LCDUI



MacDraw Reborn!



3D Cube Test in JBenchMark

Tungsten Design Advantages

High resolution screen
(320x320 pixels,16-bit color) makes text, numbers and images ultra-sharp

High capacity battery
A full day of work with wireless active

Thumb keyboard and Five-Way Nav
rapid data entry and control

Expansion cards **can add memory, software, and SD/IO devices like digital cameras**



Toolkit and Runtime Availability

WebSphere Micro Environment Toolkit for Palm OS Developers

- Available NOW at no charge through http://palm.com/java
- Includes development-licensed WebSphere Micro Environment runtime, JarToPRC utility, and support for standards-based debuggers
- Standalone utility, for use with any J2ME-compliant development environment

WebSphere Micro Environment Runtime (End-User License)

- Available soon for End-User Downloads in through Palm Support site
- No-Charge download for any Tungsten handheld purchased after release date
- Bundled with all future Palm Tungsten devices

J2ME Development Specifics

Develop, package, and test:

- Standard MIDlets
- CLDC main() applications

Features

- High-performance graphics
- HTTPS connector
- Custom application icons
- Fast PNG decoding
- Image pre-translation
- Five-way navigation

WebSphere Micro Environment Target Runtime

Two production components

- J9_vm_bundle.prc – the CLDC VM
- Midp21.prc – the MIDP/CLDC class libraries

Two debugging components

- J9_vm_debug_bundle.prc – the VM's debug module
- J9pref.prc – Palm preferences panel for configuring debug sessions

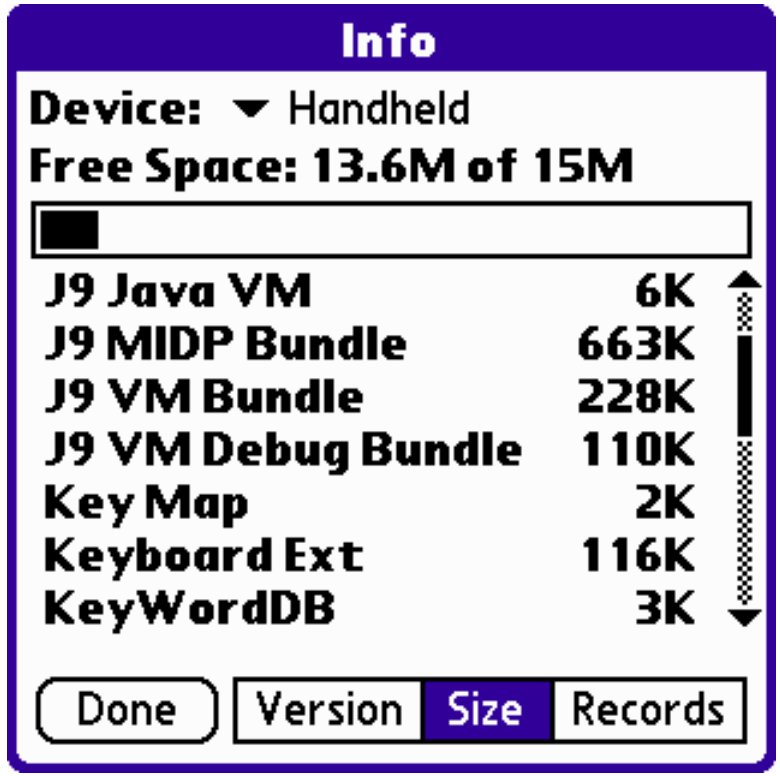
Where in the distribution?

Toolkit:

IBM/WebSphere Micro Environment/prc

Device Developer:

wsdd5.0/ive-2.1/runtimes/palmos/68k/ive/bin



Toolkit Usage With Existing IDEs

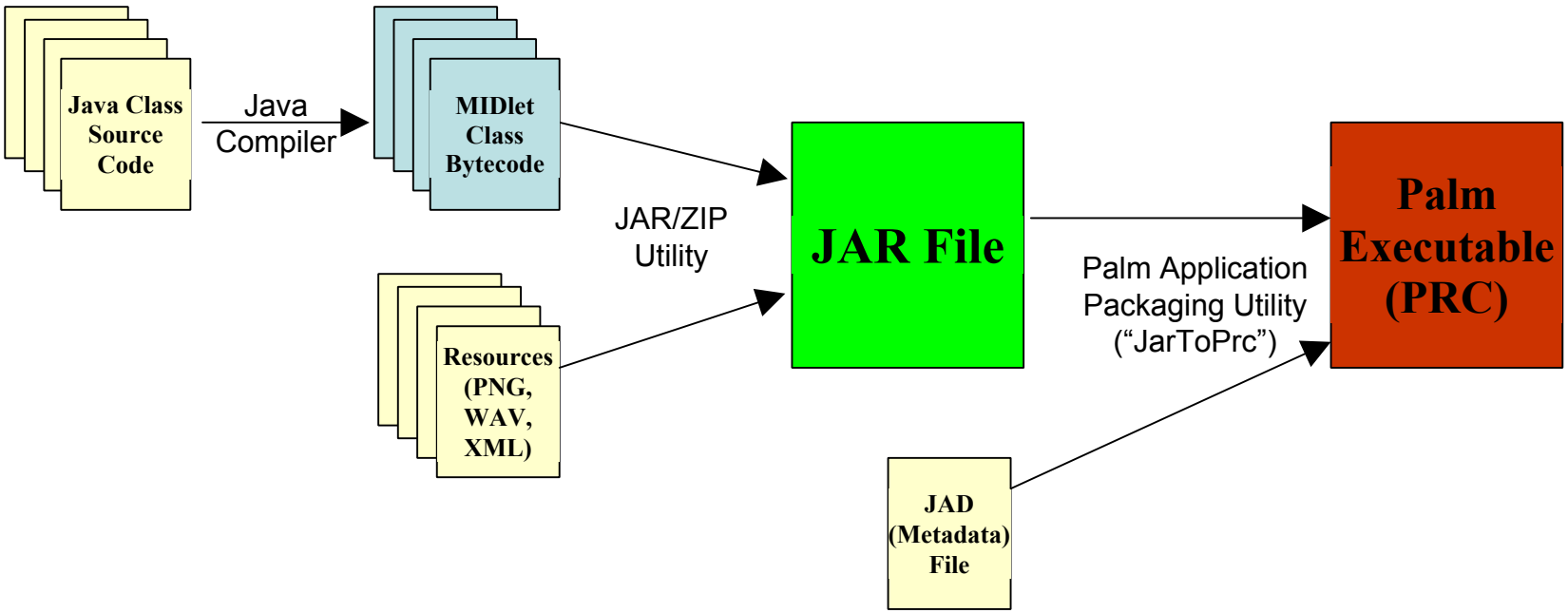
If you already have a J2ME IDE you like, use the Toolkit to easily integrate support for WebSphere Micro Environment

- Create your MIDlet suite as usual
- Add a step to the end of the automated build process to convert JAR/JAD to WebSphere PRC

Enables developers to perform two functions:

1. Package apps for WebSphere on Palm OS
2. Debug apps on WebSphere on Palm OS using standard Java debugger

Process
From .Java to .PRC

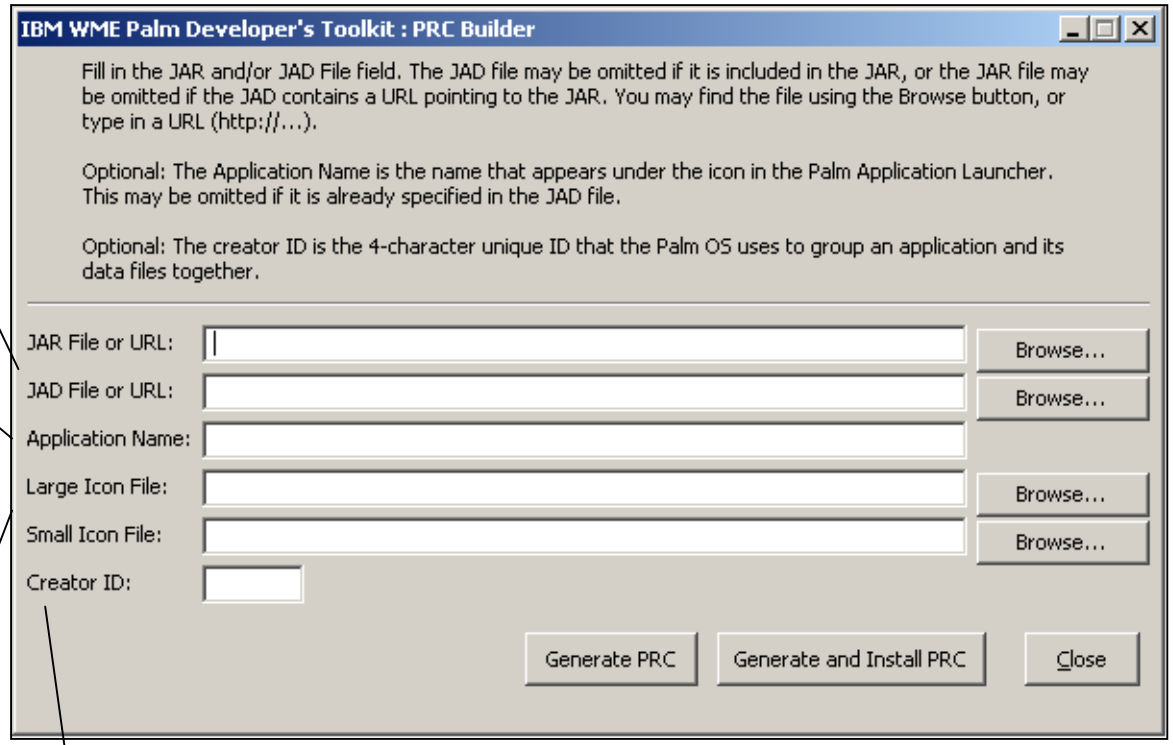


Java Application Packaging "Jar To PRC"

Standard JAR and JAD go in here

Specify a name for your Palm application, or let us find it in the JAD file

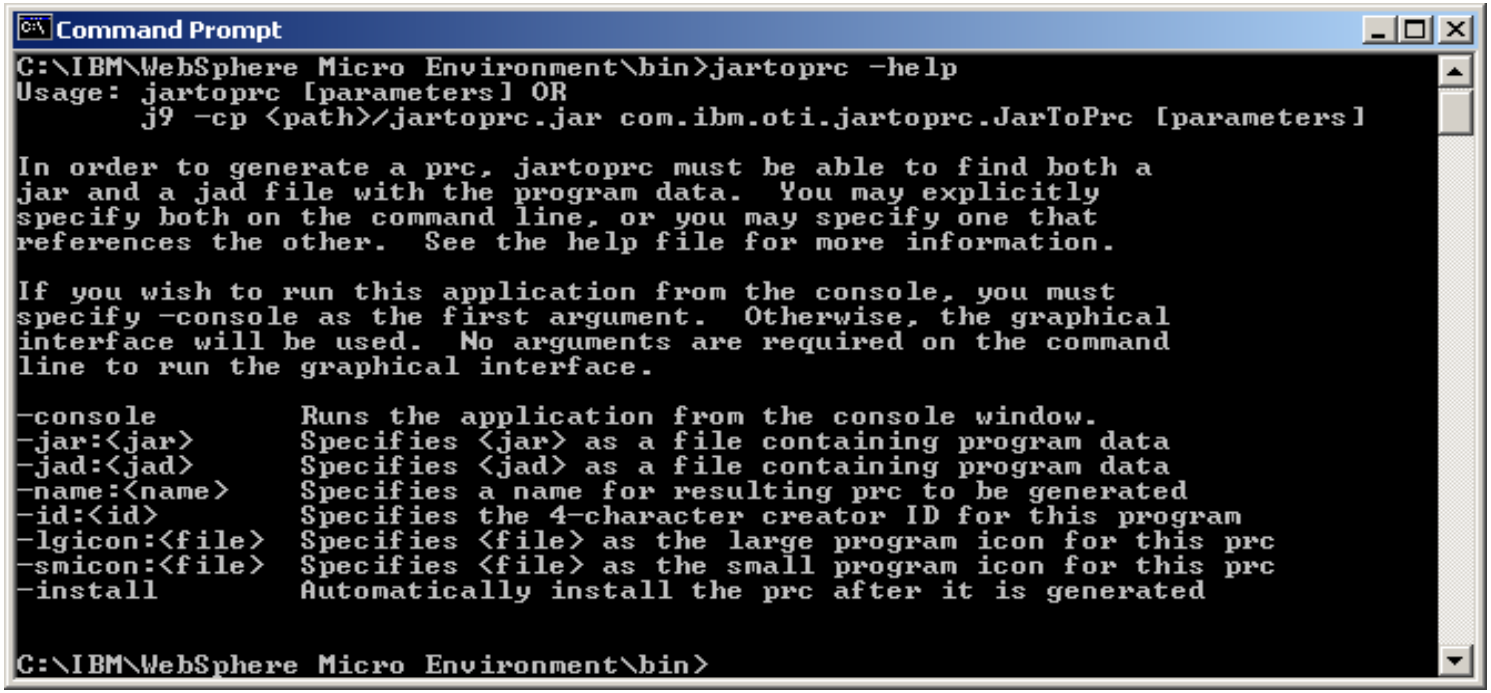
Add custom application icons for Palm Launcher icon and list views



Put your registered Creator ID here, or leave blank for testing

Java Application Packaging “Jar To PRC”

Same functionality in graphical and command-line modes:



Use command-line mode to integrate with your IDE or build-script

Java Application Packaging “Jar To PRC”

Input:

- Sample.jad
- Sample.jar

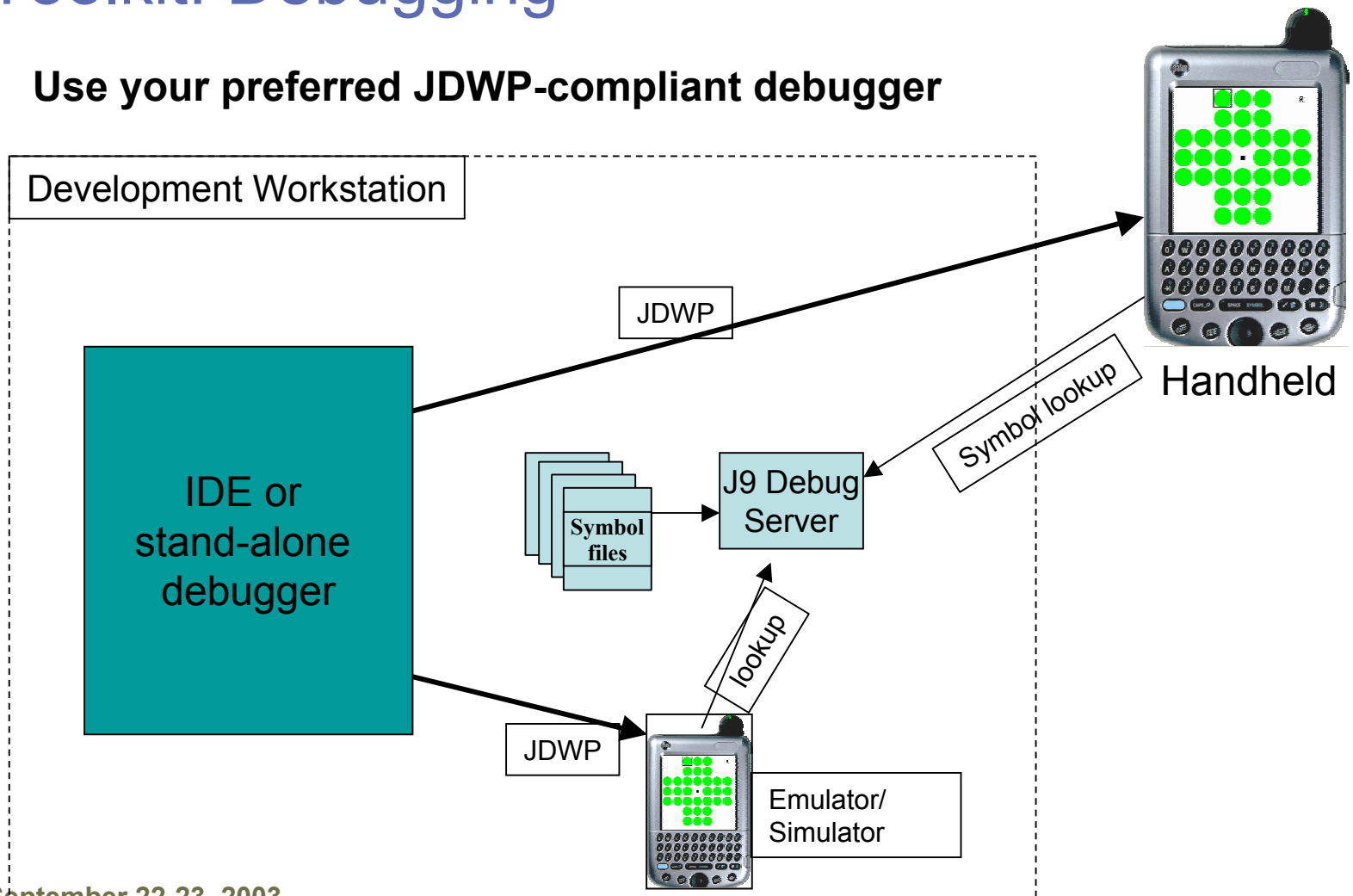
Output

- Sample.prc
- Sample.sym
- Sample1.sym
- Sample2.sym
- ….

Sample.prc: your application, ready for installation on handheld
Sample(n).sym: off-target symbol files for debugging

Toolkit: Debugging

Use your preferred JDWP-compliant debugger

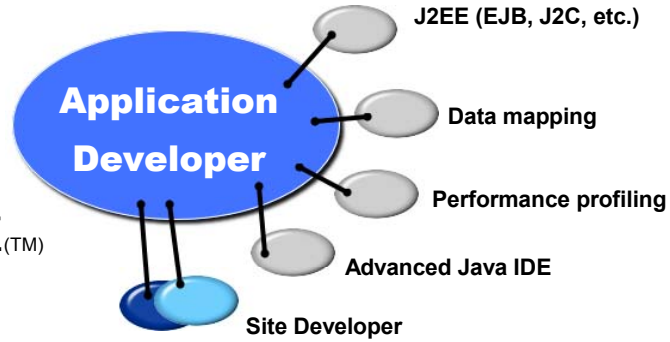


WebSphere Studio

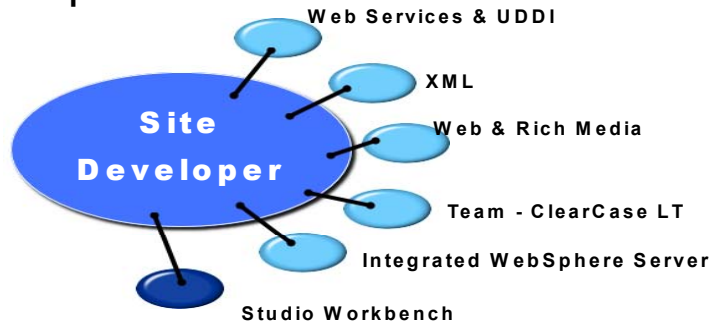


- Open standards based
  - Powered by Eclipse technology
- Productivity through integration
  - Tools plug into each other
  - IBM, OEM and ODM
- Strong version control

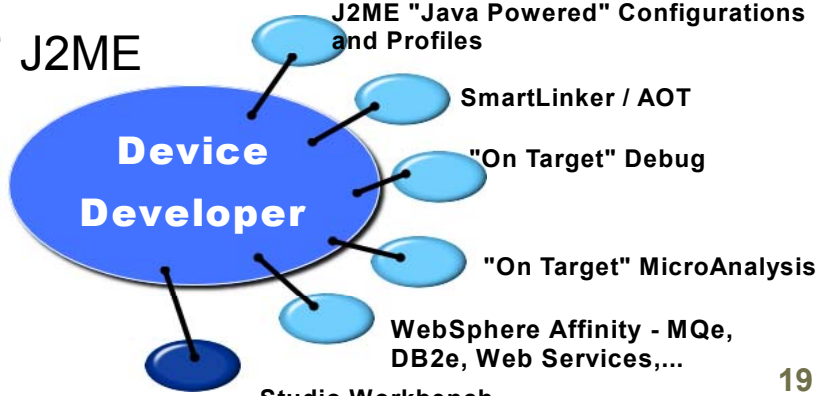
J2EE(TM)



Site Development



J2ME



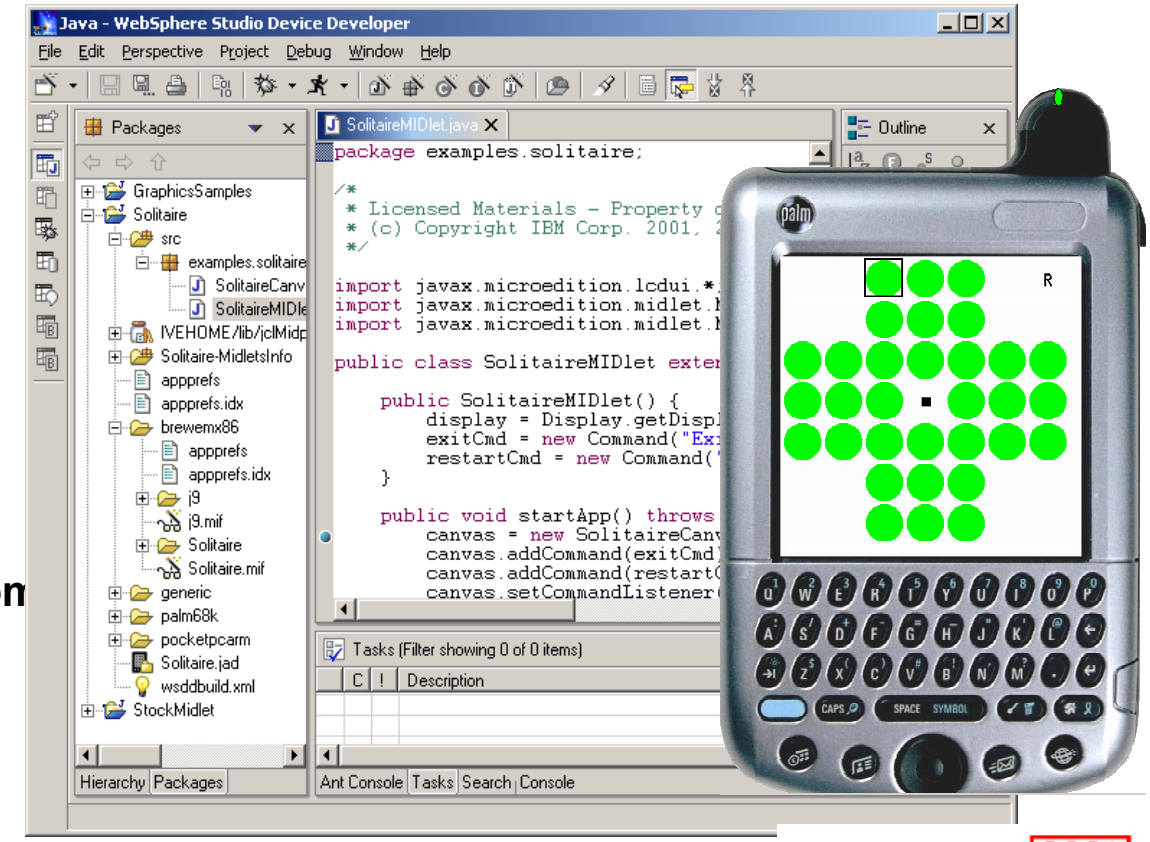
WebSphere Studio Device Developer

Complete IDE

- Create, edit
- Optimize
- Debug / Test
- Emulation
- Integrate
- Version Control I/F

Tool Plug in interface

Runtime Library updates from web



EDITORS' CHOICE JavaWorld WINNER 2002

2002 JAVAPro READER'S CHOICE

2001 JAVAPro READER'S CHOICE

Using the Toolkit

Download WebSphere Micro Environment Toolkit for Palm OS Developers:

http://palm.com/java

Read the User Guide:

- IBM/WebSphere Micro Environment/UserGuide.doc

Follow the User Guide to package, run, and debug the sample

Integrate the Toolkit with your development environment

Package, run, and debug your MIDlet!

Participate in developer community at:

- http://pluggedin.palm.com
- news://news.software.ibm.com/ibm.software.websphere.studio.device-developer

Using WSDD

Download and install Device Developer 5.5 (if you don't already have it):

http://www.ibm.com/embedded/

Use *Software Updates* feature to upgrade to 5.6 Tech Preview

- Installation instructions at http://www.ibm.com/software/wireless/wsdd/upgrades_migrations.html

Load the Solitaire example:

- File → New → Project...
  - Examples → J2ME for J9 → CLDC/MIDP → MIDP – Solitaire

Or create your own MIDlet:

- File → New → Project…
  - J2ME for J9 → Create MIDlet Suite

Create Device, Build, and Launch configurations (previous three slides)

Run and debug application!

Participate in developer community at:

- news://news.software.ibm.com/ibm.software.websphere.studio.device-developer
- http://pluggedin.palm.com

Palm Developer Resources Online

Palm's PluggedIn website (**pluggedin.palm.com**) supports developers working with Palm-branded handheld devices, and provides toolkit and device simulator downloads.

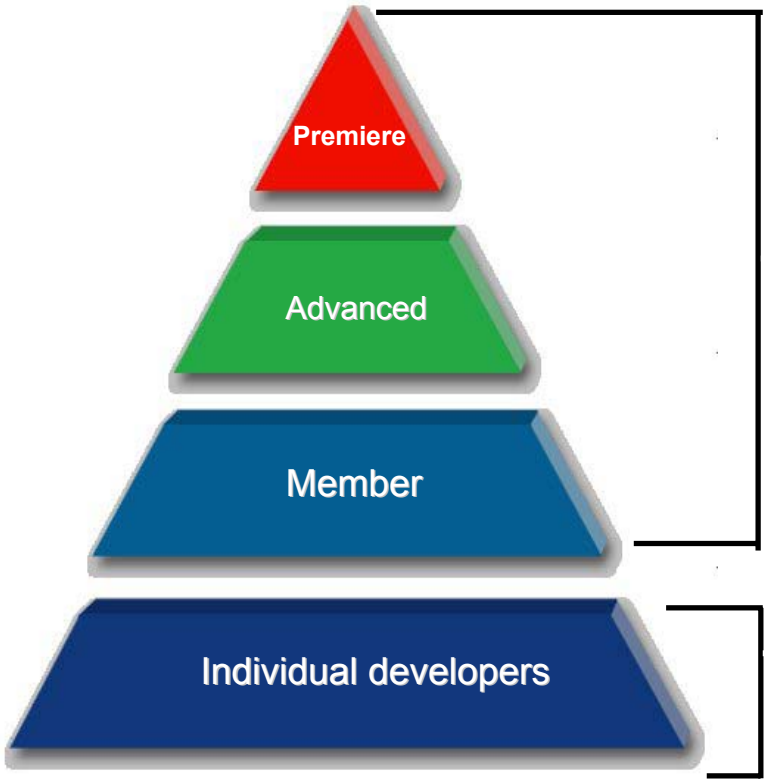
The Java section (**palm.com/java**) provides Palm or Tungsten specific Java content, articles, examples, and links.

For general Palm OS development, visit **palmsource.com/developers**

For Java developers, visit **ibm.com/developerWorks**

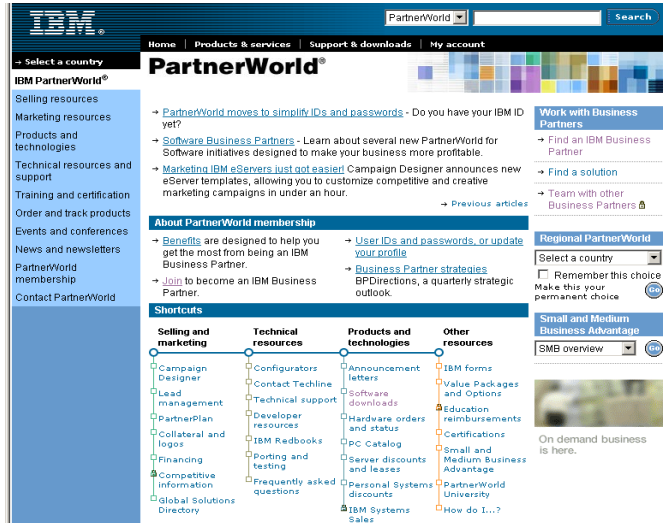


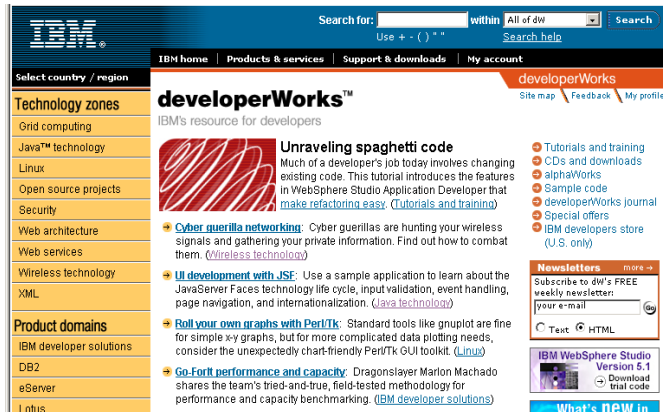
Developer Resources From IBM



Business Partners

Individuals





developerWorks Resources

IBM's technical resource for individual developers:

- **Tutorials**
- **Articles**
- **Tools**
- **Development tips**
- **White papers**
- **Forums**
- **Downloads**
- **developerWorks Journal newsletter ***



Technology info:

- Java
- Linux
- Web development
- Wireless
- XML

IBM Product info:

- WebSphere
- DB2
- Lotus
- Tivoli
- Rational

* Developers can contribute articles & earn money!

PartnerWorld for Developers

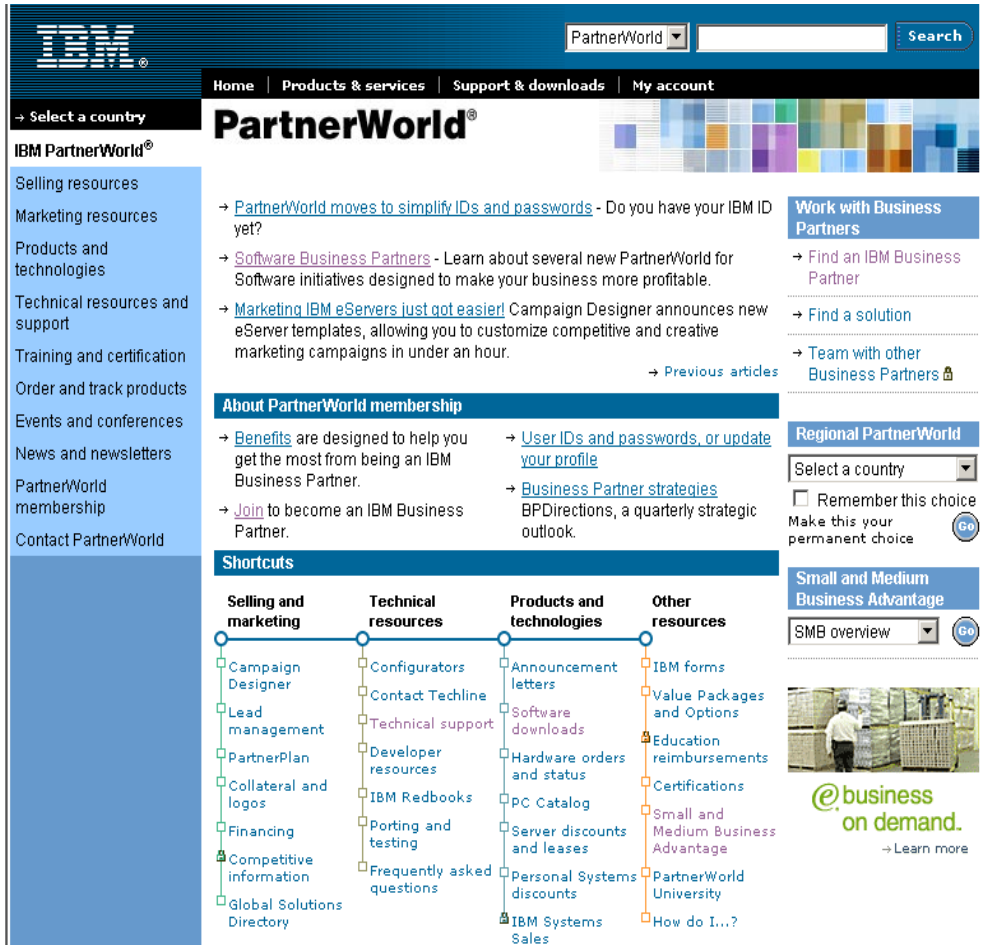
IBM's technical resource for commercial solution developers.

Objectives:

- Faster time to market
- Lower cost of doing business
- Broader business opportunities

Benefits & offerings:

- Marketing & sales
- Technical support
- Education & training
- Business incentives
- Financing
- Relationship management



IBM Programs for Palm Developers

WebSphere Studio Device Developer Workshops

- 3-day hands-on sessions, WW venues, free to qualified partners
- Learn Eclipse-based tools, WebSphere Micro Environment, WebSphere Programming Model
- New labs target Java Midlets to run on Palm Tungsten devices
- More information at: www.ibm.com/software/pervasive/education/workshop_a.shtml

Visibility on IBM Pervasive Computing website

- Opportunity to showcase Palm MIDP applications
- Links to Palm developer websites for additional information, demo's, etc.

Exposure to IBM Pervasive Sales

- IBM "Lunch & Learn" sessions preview partner companies/applications
- Opportunity to focus on enterprise-enabling Palm MIDP applications
- Potential for direct contact with IBM Pervasive Sales

IBM Pervasive Computing Embedded Summit for Business Partners

- Invitation-only event in advance of Embedded System Conference
- Meet the embedded JVM and tools developers
- Network with other embedded developers using IBM device software

Future Events and Plans

Palm Solutions Group and Handspring merging to become palmOne (pending shareholder approval)

Supporting J2ME application developers through PluggedIn program and Java developer and marketing events

- Developer events in San Jose and New York in October

Working internally and with partners to expand Java support

- PDA Optional Extensions, Bluetooth, Web Services, Wireless Messaging, Mobile Media….
- Other IDE vendors to integrate toolkit

Continuing to plan our long-term Java roadmap

- New devices, New OS, New JSRs (CDC, Personal Profile)

Web Resources

- Palm Java information www.palm.com/java
- Palm OS Information www.palmsource.com/developers
- Palm PluggedIn developer's program http://pluggedin.palm.com
- IBM WebSphere Micro Environment
  www.ibm.com/software/wireless/wme
- IBM WebSphere Studio Device Developer
  www.ibm.com/embedded
- WebSphere Studio Device Developer newsgroup
  news://news.software.ibm.com/ibm.software.websphere.studio.device-developer
- IBM Pervasive Computing www.ibm.com/pvc
- IBM developerWorks www.ibm.com/developerworks
- IBM PartnerWorld www.ibm.com/partnerworld
- IBM PartnerWorld, Pervasive
  www.developer.ibm.com/pvc/index.html
- IBM PartnerWorld for Developers (registration)
  www.developer.ibm.com/member/register/registration.html

Danke! Merci! Grazie! Obrigado! Thanks!

Please send questions or feedback to:

Nathanial.Freitas@corp.palm.com

IBM WebSphere® Micro Environment

Java POWERED

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